EXPERIMENT-5 Develop a customized ping command to test the server connectivity

Aim - To develop a customized Ping program in Python that tests server connectivity by sending ICMP echo requests and displaying the response time and status.

Requirements

Hardware

• Laptop/Desktop

• Internet or Local Area Network (LAN)

Software

• Python 3.8 or above

• Visual Studio Code (VS Code) / IDLE / PyCharm

• Windows 10/11 or Ubuntu Linux.

custom_ping.py - C:/Users/MAHATHI A J/Documents/CustomPing/custom_ping.py (3.14.0)

File Edit Format Run Options Window Help

```
import subprocess
import platform

host = input("Enter the server name or IP address: ")
if platform.system().lower() == "windows":
    command = ["ping", "-n", "4", host]
else:
    command = ["ping", "-c", "4", host]

print("\nTesting connectivity...\n")

result = subprocess.run(
    command,
    capture_output=True,
    text=True
)

print(result.stdout)

if result.returncode == 0:
    print("Server is Reachable.")
else:
    print("Server is Not Reachable.")
```

IDLE Shell 3.14.0

File Edit Shell Debug Options Window Help

```
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct  7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
======= RESTART: C:/Users/MAHATHI A J/Documents/CustomPing/custom_ping.py ======
Enter the server name or IP address: google.com

Testing connectivity...


Pinging google.com [142.250.67.46] with 32 bytes of data:
Reply from 142.250.67.46: bytes=32 time=19ms TTL=119
Reply from 142.250.67.46: bytes=32 time=22ms TTL=119
Reply from 142.250.67.46: bytes=32 time=26ms TTL=119
Reply from 142.250.67.46: bytes=32 time=22ms TTL=119

Ping statistics for 142.250.67.46:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 19ms, Maximum = 26ms, Average = 22ms

Server is Reachable.
>>>
======= RESTART: C:/Users/MAHATHI A J/Documents/CustomPing/custom_ping.py ======
Enter the server name or IP address: 192.168.1.100

Testing connectivity...


Pinging 192.168.1.100 with 32 bytes of data:
Request timed out.
Request timed out.

Ping statistics for 192.168.1.100:
    Packets: Sent = 2, Received = 0, Lost = 2 (100% loss),

Server is Not Reachable.
>>>
```